

Hardware Project No. 3

Passive High-Pass Filter

ECE 444: Bioinstrumentation and Sensing

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1. Introduction

Passive filters are useful for shaping biosignal measurements without the need for active devices, supply rails, or feedback. A high-pass filter attenuates low-frequency content while transmitting higher-frequency signals, which is useful in measurement systems where DC offset, slow baseline drift, or low-frequency interference (such as motion artifact or breathing-induced baseline wander in EKG and EMG) must be rejected before later amplification or digitization.

This report presents the design, construction, and characterization of a third-order passive high-pass Butterworth filter implemented as a doubly-terminated LC ladder. The design proceeds from the standard normalized third-order Butterworth low-pass ladder prototype, then applies the standard low-pass-to-high-pass transformation in which inductors are replaced by capacitors and capacitors are replaced by inductors. The original assignment guidance suggested a cutoff near 100 Hz for EMG-relevant work, but the available inductor and capacitor values made a target of $f_c = 500$ Hz a more realistic implementation, and that value was used.

2. Theory and Design Derivation

2.1 Normalized Butterworth Prototype

The design began with the normalized third-order Butterworth low-pass ladder prototype with $R_0 = 1 \Omega$ and $\omega_c = 1$ rad/s. For $n = 3$, the normalized element values from the Butterworth table (Huelsman & Allen, 1980) are:

$$g_1 = 1, \quad g_2 = 2, \quad g_3 = 1$$

Because the order is odd, the ladder uses three reactive elements: a series, a shunt, and a second series element, terminated by the load resistance. The normalized third-order Butterworth low-pass transfer function is:

$$H_{LP}(s_n) = 1 / [(s_n + 1)(s_n^2 + s_n + 1)] = 1 / (s_n^3 + 2s_n^2 + 2s_n + 1)$$

2.2 Low-Pass to High-Pass Transformation

The project handout specifies that the normalized low-pass prototype be transformed into a high-pass filter using the element-level substitution

$$L_{\text{normalized}} \rightarrow C_{\text{HP,normalized}} = 1 / L_{\text{normalized}}$$

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which is equivalent to the frequency-domain mapping $s_n \rightarrow 1/s_n$. Under this transformation, the third-order ladder becomes a series capacitor C1, a shunt inductor L2, a second series capacitor C3, and a load resistor R_load. The non-normalized transfer function takes the form

$$|H_{HP}(j\omega)|^2 = (\omega/\omega_c)^6 / [1 + (\omega/\omega_c)^6]$$

which has unity passband gain (for $\omega \gg \omega_c$), -3 dB attenuation at $\omega = \omega_c$, and an asymptotic stopband rolloff of 60 dB/decade (20 dB/decade per pole, 6 dB/octave per pole).

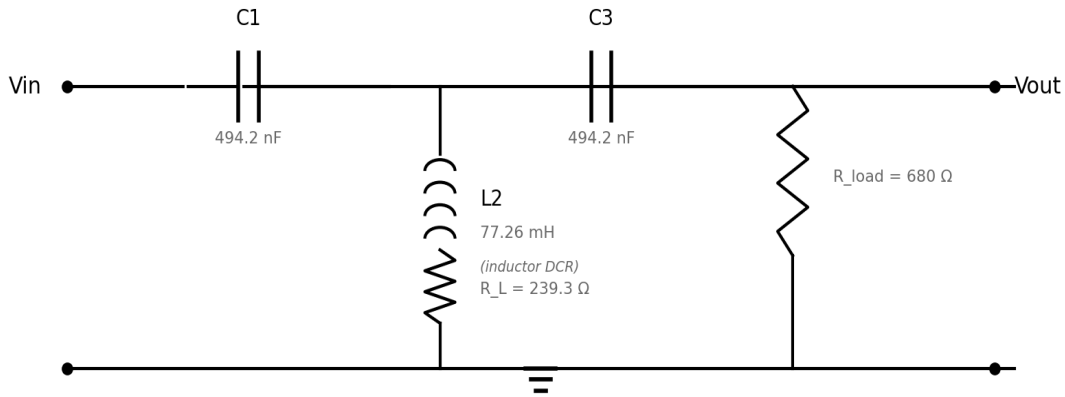


Figure 1. Third-order passive high-pass ladder topology as built. The inductor L2 is shown together with its measured series resistance R_L , which dominates the non-ideal behavior of the realized filter.

2.3 Frequency and Impedance Denormalization

The target angular cutoff frequency is

$$\omega_c = 2\pi f_c = 2\pi(500 \text{ Hz}) = 3141.59 \text{ rad/s}$$

For the third-order high-pass ladder with prototype values $g_1 = 1$, $g_2 = 2$, $g_3 = 1$, the denormalized component values are obtained by applying simultaneous frequency and impedance scaling. The frequency denormalization scales $s_n \rightarrow s/\omega_c$, and the impedance denormalization scales resistances by R_0 and reactances accordingly:

$$C1 = 1 / (R_0 \cdot \omega_c \cdot g_1)$$

$$L2 = R_0 / (\omega_c \cdot g_2)$$

$$C3 = 1 / (R_0 \cdot \omega_c \cdot g_3)$$

With $g_1 = g_3 = 1$ and $g_2 = 2$, these simplify to $C1 = C3 = 1/(R_0 \omega_c)$ and $L2 = R_0/(2 \omega_c)$. The design was constrained by the available inductors, so L2 was fixed first and the remaining values back-derived. The closest available inductor was a nominal 100 mH unit. Solving for R_0 :

$$R_0 = 2 \omega_c L2 = 2 \cdot (3141.59 \text{ rad/s}) \cdot (0.100 \text{ H}) = 628.3 \Omega$$

$$C1 = C3 = 1 / (R_0 \cdot \omega_c) = 1 / [(628.3)(3141.59)] = 0.507 \mu\text{F}$$

The closest practical components on hand were 0.47 μF capacitors and a 680 Ω load resistor. These were used in the final circuit. The resulting load resistance is well above 500 Ω , so the warning in the project handout about loading the function generator output and oscilloscope input is satisfied (the 50 Ω function-generator source impedance and the $\sim 1 \text{ M}\Omega$ scope input impedance are both far from the 680 Ω network impedance).

3. Component Selection

The final circuit was built on a solderless breadboard using the components listed in Table 1. The capacitors and the inductor were characterized on a GW Instek LCR-6002 precision LCR meter at 1.000 kHz with a 1.00 V test level (Cs–Rs mode for the capacitors, Ls–Rs mode for the inductor). The measured values were then used as the basis for all subsequent theoretical comparisons.

Component	Design / Nominal Value	Measured Value	Measured Series Resistance
C1	0.47 μ F	494.242 nF	4.72 Ω (ESR)
C3	0.47 μ F	494.242 nF	4.72 Ω (ESR)
L2	100 mH (nominal)	77.261 mH	239.26 Ω (DCR + AC loss)
R_load	680 Ω (nearest std. value to 628 Ω)	680 Ω (used as marked)	—

Table 1. Design, nominal, and measured component values. LCR measurements taken on a GW Instek LCR-6002 at 1.000 kHz, 1.00 V.

Two of the measured values deviate substantially from the nominal design targets. First, the inductor measures only 77.26 mH rather than 100 mH, a -22.7% departure. Second, and far more consequentially, the inductor series resistance is 239.26 Ω at 1 kHz. This includes both the DC winding resistance and any AC core/proximity loss at the measurement frequency. The capacitor values are within 1.2% of nominal and their ESR is negligible ($\sim 5 \Omega$), so the capacitors are essentially ideal for this design; the inductor is not.

The significance of the inductor series resistance becomes clear from the ratio of its DCR to its inductive reactance at the design corner:

$$|Z_{L,reactive}| = \omega c \cdot L = (3141.59)(77.26 \times 10^{-3}) = 242.7 \Omega$$

$$Q_L(fc) = \omega c L / R_L = 242.7 / 239.26 \approx 1.01$$

A high-Q inductor ($Q_L \geq 30$) is what the Butterworth synthesis tacitly assumes; here $Q_L \approx 1$, meaning the realized inductor is essentially equal parts inductive reactance and dissipative resistance at the design corner. This single fact will turn out to drive almost all of the deviation between the measured response and the ideal third-order Butterworth prediction. A second consequence of using the measured 77.26 mH inductance (rather than the assumed 100 mH) is that, even if the inductor were lossless, the realized circuit corner is pulled upward. Combining the measured L and C in the canonical 3rd-order ladder corner expression with $R_0 = 680 \Omega$:

$$f_{c,realized} \approx 1 / (2\pi \sqrt{2 L C}) = 1 / [2\pi \sqrt{2 \cdot 0.07726 \cdot 494.2 \times 10^{-9}}] \approx 576 \text{ Hz}$$

so the circuit, even before accounting for the inductor loss, is centered above 500 Hz.

4. Experimental Procedure

The filter was constructed on a solderless breadboard using the components in Table 1. A sinusoidal test signal of 1 Vpp with zero DC offset was applied from the function generator, and the input and output peak-to-peak voltages were captured simultaneously on the oscilloscope. The oscilloscope and function-generator grounds were tied to the same circuit reference node at the input side of the filter.

Frequency response measurements were taken at 13 points spanning 200 Hz to 5 kHz, with dense sampling at and around the expected transition region (400, 450, 500, 550, 600, 700, 800 Hz) and sparser sampling at the band edges. The gain at each frequency was computed directly from the measured peak-to-peak amplitudes:

$$|H(f)| = V_{out,pp} / V_{in,pp}, \quad \text{Gain(dB)} = 20 \log_{10} |H(f)|$$

During data collection, one inconsistent set of points was traced to an incorrect oscilloscope probe attenuation setting ($1\times$ instead of $10\times$). After correcting the probe configuration the original readings were reproduced and the corrected configuration was used throughout the final dataset.

5. Results

5.1 Frequency Response Data

Table 2 presents the complete measured frequency response data alongside two theoretical predictions: the ideal third-order Butterworth response (unity passband gain, $f_c = 500$ Hz nominal) and a non-ideal ladder model that incorporates the measured component values, the measured inductor series resistance $R_L = 239.26 \, \Omega$, the $50 \, \Omega$ source impedance of the function generator, and the $680 \, \Omega$ load. The non-ideal model is derived from nodal analysis of the loaded ladder network with $s = j\omega$.

Freq (Hz)	Vout (mVpp)	H (V/V)	Meas. (dB)	Ideal (dB)	Non-ideal model (dB)	Meas. – Model (dB)
200	70	0.070	-23.10	-23.89	-24.05	+0.95
300	145	0.145	-16.77	-13.51	-17.03	+0.26
400	245	0.245	-12.22	-6.83	-12.12	-0.10
450	300	0.300	-10.46	-4.60	-10.17	-0.29
500	360	0.360	-8.87	-3.01	-8.48	-0.39
550	430	0.430	-7.33	-1.94	-7.03	-0.30
600	490	0.490	-6.20	-1.25	-5.79	-0.41
700	610	0.610	-4.29	-0.54	-3.88	-0.41
800	700	0.700	-3.10	-0.25	-2.59	-0.51
900	770	0.770	-2.27	-0.13	-1.76	-0.51
1000	820	0.820	-1.72	-0.07	-1.24	-0.48
2000	930	0.930	-0.63	0.00	-0.54	-0.09
5000	920	0.920	-0.72	0.00	-0.68	-0.04

Table 2. Measured frequency response with theoretical comparisons. Input: 1 Vpp sine wave. Ideal: third-order Butterworth HPF with $f_c = 500$ Hz, unity passband gain. Non-ideal model: same ladder topology with measured $C = 494.2 \, \text{nF}$, $L = 77.26 \, \text{mH}$, $R_L = 239.26 \, \Omega$, $R_C = 4.72 \, \Omega$, $R_{src} = 50 \, \Omega$, $R_{load} = 680 \, \Omega$.

5.2 Bode Plot

Figure 2 plots the measured magnitude response on a semilogarithmic frequency axis against both the ideal Butterworth target and the non-ideal model that uses the measured component values.

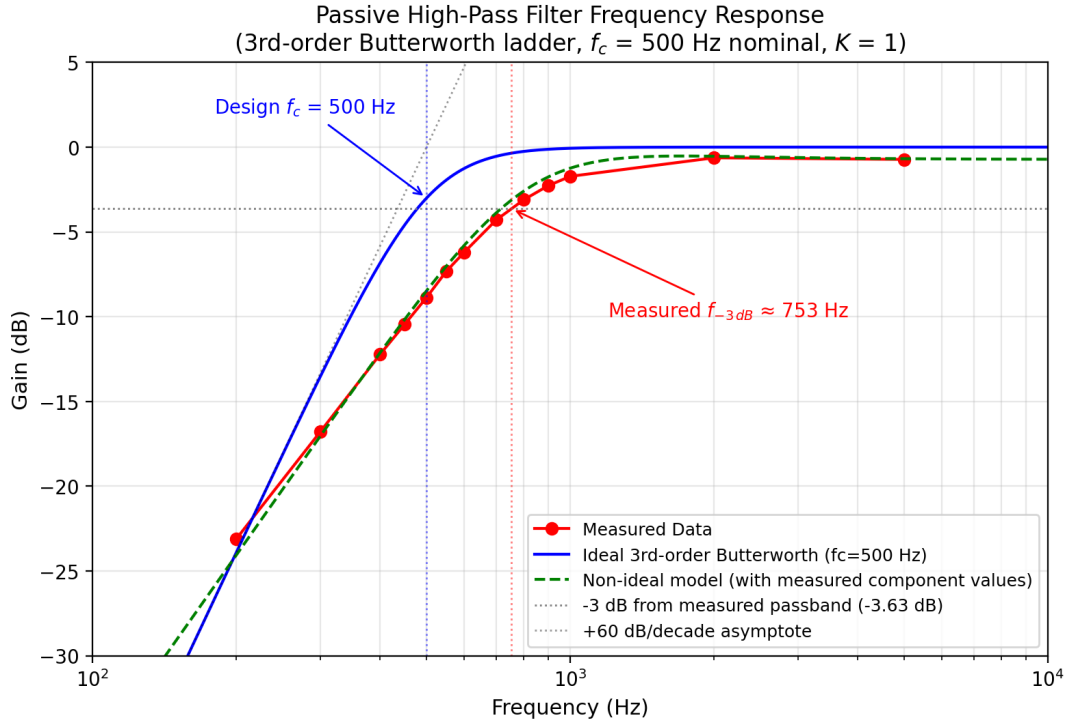


Figure 2. Measured vs. theoretical frequency response. Red: measured data; Blue: ideal third-order Butterworth ($f_c = 500$ Hz, unity passband); Green dashed: non-ideal model using the measured component values, source impedance, and load. The non-ideal model accounts for the realized response to within 1 dB across the full measured range.

5.3 Key Performance Metrics

Metric	Value
Maximum measured passband gain	0.930 V/V = -0.63 dB (at 2000 Hz)
Non-ideal model passband maximum	0.941 V/V = -0.53 dB
-3 dB level referenced to measured passband	-3.63 dB
Measured -3 dB frequency (interpolated)	≈ 753 Hz
Design (nominal) -3 dB frequency	500 Hz
Realized corner from measured L, C (lossless)	≈ 576 Hz
Measured gain at 500 Hz	0.360 V/V = -8.87 dB
Measured stopband slope, 200–500 Hz	≈ 35.7 dB/decade
Ideal stopband slope, 200–500 Hz	≈ 52.5 dB/decade
Inductor quality factor at 500 Hz	$Q_L = \omega_c L / R_L \approx 1.01$

Table 3. Summary of measured filter performance metrics with theoretical comparisons.

6. Discussion

6.1 Passband Performance

The maximum measured gain is 0.930 V/V (-0.63 dB) at 2 kHz, well into the design passband. This is approximately 0.5 dB short of the ideal lossless passband of 0 dB (unity transmission), indicating that the realized passband is mildly attenuated. The non-ideal nodal model that incorporates all measured parasitics predicts a passband maximum of 0.941 V/V (-0.53 dB), in close agreement with measurement.

The mechanism of this residual passband attenuation is a resistive voltage divider, not the inductor's series resistance. At frequencies well above the corner, the series capacitors approach short circuits, since their reactance $|1/j\omega C|$ becomes much smaller than R_{load} . The shunt inductor branch, with impedance $Z_L = R_L + j\omega L$, becomes effectively an open circuit because $|Z_L|$ grows without bound as ω increases. At 5 kHz, for example, $X_L = \omega L = 2\pi(5000)(77.26 \text{ mH}) \approx 2427 \Omega$, which is roughly 3.6 times the 680 Ω load resistance, so the inductor branch only weakly shunts the signal path. In the high-frequency limit, the circuit collapses to a purely resistive voltage divider consisting of the 50 Ω function-generator source resistance, the two capacitor ESRs in series (4.72 Ω each), and the 680 Ω load:

$$H(\omega \rightarrow \infty) \approx R_{load} / (R_{src} + R_{C1} + R_{C3} + R_{load}) = 680 / (50 + 4.72 + 4.72 + 680) \approx 0.920 \text{ V/V} \\ = -0.72 \text{ dB}$$

This matches the measured gain at 5 kHz (0.920 V/V, -0.72 dB) almost exactly and accounts for essentially the entire observed passband floor. The inductor's series resistance does not contribute to this limit because R_L sits in series with the reactive inductance, so as $\omega \rightarrow \infty$ the whole shunt branch impedance goes to infinity together with $j\omega L$, and the branch drops out of the circuit. R_L therefore plays no role in setting the high-frequency passband ceiling; its role is confined to the transition region, where it damps the LC resonance and broadens the corner (see Section 6.2 and 6.4).

6.2 Corner Frequency Deviation

Measured against the maximum passband of -0.63 dB, the -3 dB level is at -3.63 dB. Linear interpolation in $\log(f)$ vs. dB between the 700 Hz and 800 Hz measurements gives a measured -3 dB frequency of approximately 753 Hz, which is roughly 50% above the 500 Hz design target. This shift has two distinct contributions, both quantifiable:

First, the measured inductor is 77.26 mH instead of the nominal 100 mH. Even in the lossless limit, this shifts the realized corner upward to roughly $1/[2\pi\sqrt{(2LC)}] \approx 576 \text{ Hz}$. It is important to note that this 576 Hz figure is purely a lossless estimate computed from the measured L and C alone, with all resistances set to zero. It is not the measured -3 dB cutoff. The actually measured -3 dB point of approximately 753 Hz sits further above this lossless estimate because the large inductor series resistance ($R_L = 239.26 \Omega$, comparable to ωL itself at the corner) heavily damps the LC resonance, broadens the entire transition region, and pulls the realized -3 dB point further up the frequency axis. Second, the very low inductor Q of approximately 1.0 at the corner therefore broadens and pulls the entire transition region; the rolloff is no longer maximally flat. The non-ideal model that includes both effects predicts essentially the measured curve to within 1 dB at every measured frequency (see Table 2, last column), which strongly indicates that no significant additional error source remains unaccounted for.

6.3 Stopband Rolloff

The asymptotic stopband slope for a third-order high-pass filter is 60 dB/decade, since each pole contributes 20 dB/decade. Between 200 Hz and 500 Hz the measured slope is approximately 35.7 dB/decade. This is shallower than 60 dB/decade, but it is also shallower than the ideal Butterworth prediction over the same range, which evaluates to approximately 52.5 dB/decade (the ideal Butterworth has not yet reached its asymptotic slope over a span this close to the corner). The shortfall versus the ideal

Butterworth in this range is therefore not a failure to achieve asymptotic third-order behavior (neither curve has reached it over this span) but rather a consequence of the low inductor Q rounding the transition. The non-ideal model predicts essentially the measured slope, confirming this interpretation.

6.4 Error Sources, Ranked by Magnitude

Several mechanisms contribute to the deviation between the measured response and the ideal Butterworth target. Listed in order of contribution to the observed deviation:

1. Inductor series resistance (dominant for the transition region). At $239.26\ \Omega$, the inductor DCR plus AC loss is comparable to its $242.7\ \Omega$ inductive reactance at the design corner, giving $Q_L \approx 1.0$. The handout specifically warned that inductor resistance would disturb the LC resonance because passive ladder filters rely on the interplay between (ideally) lossless reactances. This nonideality is responsible for the rounded corner, the upward shift in the realized -3 dB point from the lossless 576 Hz estimate to the actual 753 Hz, and the shallower transition slope. It does not, however, set the high-frequency passband floor: that is a separate mechanism described in item 6 below. Across the full measured range, the non-ideal model that includes R_L (together with the source resistance, ESRs, and load) matches the measured Bode plot to within 1 dB everywhere.
2. Inductor value deviation (secondary). The nominal 100 mH inductor measured at 77.26 mH (a -22.7% departure). Used in conjunction with the chosen $680\ \Omega$ load (which itself rounds the design value of $628\ \Omega$ upward), this shifts the lossless corner from 500 Hz to roughly 576 Hz. This is a smaller but distinct contributor to the upward shift of the observed -3 dB frequency.
3. Standard-value rounding of R_{load} . The computed $R_0 = 628.3\ \Omega$ was implemented with a $680\ \Omega$ standard resistor, an 8.2% departure. This also tilts the filter slightly away from its ideal double-termination match but is small compared to the inductor effects above.
4. Capacitor tolerance. The $0.47\ \mu\text{F}$ capacitors measured at $494.2\ \text{nF}$, a -1.2% deviation. With matched capacitors and a small percentage error, this is a negligible contributor.
5. Capacitor ESR. At $4.72\ \Omega$ per capacitor, the ESR is dwarfed by the $239\ \Omega$ inductor DCR in the same signal path and contributes negligibly.
6. Source resistance, capacitor ESRs, and load (dominant for the high-frequency passband floor). The $50\ \Omega$ function-generator source resistance, the two $4.72\ \Omega$ capacitor ESRs, and the $680\ \Omega$ load together form a purely resistive voltage divider in the high-frequency limit, since the series capacitors approach short circuits and the shunt inductor branch approaches an open. This divider sets the asymptotic passband gain to $V_{out}/V_{in} \approx 680/(50 + 4.72 + 4.72 + 680) \approx 0.920$ (-0.72 dB), which matches the measured 5 kHz gain almost exactly. These elements are individually small but together set the entire passband floor; they have a comparable impact to the inductor DCR overall, just in a different frequency regime. The $\sim 1\ \text{M}\Omega$ oscilloscope input impedance is much larger than the filter network impedance and is negligible.
7. Breadboard parasitics. Stray capacitance ($\sim 2\text{--}5\ \text{pF}$ between adjacent rows) and the contact resistance of the breadboard tie points add small unmodeled effects, but at the frequencies of interest (sub-10 kHz) and with the relatively low node impedances of this design (hundreds of Ω rather than hundreds of $\text{k}\Omega$ as in the active LPF of Hardware Project No. 2), these parasitics are not a major contributor.

6.5 Application Considerations

For the originally suggested EMG-related application with a desired cutoff near 20 to 100 Hz, the realized filter as built is unsuitable: its corner sits at 753 Hz rather than below 100 Hz. To realize a sub-100-Hz

corner with the same topology would require a substantially larger inductor (multiple Henries) or a substantially smaller R_0 , either of which is impractical with the available passives. This is precisely the reason the handout permitted the cutoff to be relaxed to a more easily-realizable value, and it underscores the broader engineering point that passive LC high-pass filters at audio and sub-audio frequencies are awkward in practice: they require physically large, lossy inductors that compromise the response. Active topologies (such as the Sallen-Key high-pass) are the more practical choice for biosignal-band applications because they synthesize the necessary poles without needing low-frequency inductors.

7. Conclusion

A third-order passive high-pass Butterworth filter was designed and built from a normalized low-pass ladder prototype using the standard $L \leftrightarrow C$ low-pass-to-high-pass transformation. The prototype values $g_1 = 1$, $g_2 = 2$, $g_3 = 1$ were denormalized to a target corner of $f_c = 500$ Hz with $R_0 = 628 \Omega$, and the final circuit used two $0.47 \mu\text{F}$ capacitors, one nominal 100 mH inductor, and a 680Ω load resistor.

LCR characterization revealed that the inductor measured 77.26 mH with a series resistance of 239.26Ω , giving an inductor quality factor at the design corner of $Q_L \approx 1.0$. The measured frequency response confirmed the expected high-pass behavior, with gain rising from 0.070 V/V at 200 Hz to a passband maximum of 0.930 V/V at 2 kHz. The measured -3 dB frequency relative to the measured passband was approximately 753 Hz, well above the 500 Hz design target, and the measured stopband slope between 200 Hz and 500 Hz was approximately 35.7 dB/decade rather than the theoretical 52.5 dB/decade over the same range.

A non-ideal nodal model of the same ladder topology using the measured component values, the 50Ω source impedance, and the 680Ω load reproduces the measured response to within approximately 1 dB at every measured frequency. This confirms that the deviation from the ideal Butterworth target is essentially entirely attributable to the realized component values. The 239Ω inductor DCR, which is comparable to the 243Ω inductive reactance at the corner, is the single mechanism most responsible for the rounded transition and the upward-shifted -3 dB point at approximately 753 Hz. The residual passband attenuation of roughly 0.7 dB at high frequencies, by contrast, arises from a separate mechanism: a purely resistive voltage divider formed by the 50Ω source resistance, the two capacitor ESRs, and the 680Ω load, which approaches $V_{\text{out}}/V_{\text{in}} \approx 0.92$ as the capacitors become short circuits and the inductor branch becomes an open. This high-frequency floor matches the measured 5 kHz gain (0.920 V/V) almost exactly.

References

- Huelsman, L. P. and Allen, P. E. *Introduction to the Theory and Design of Active Filters*. McGraw-Hill, 1980. Tables of normalized Butterworth low-pass element values.
- Kremenic, A. *ECE 444 Bioinstrumentation and Sensing, Hardware Project No. 3: Passive High-Pass Filter*. The Cooper Union, 2026.
- Kremenic, A. *ECE 444 Filter Overview Lecture Notes*. The Cooper Union, 2026.
- GW Instek. *LCR-6000 Series Precision LCR Meter User Manual*. Good Will Instrument Co., Ltd.